

# APDTIA4B-2.5G-W

## OVERVIEW

Avalanche photodiode with a low-noise transimpedance amplifier with auto gain control coupled to an optical fiber and packaged into a hermetic case.

## MAIN FEATURES

- Operation wavelength 1260 – 1640 nm
- Data rate: 2.5 Gbps
- Sensitivity: -35 dBm
- Package types: coaxial with or without bracket
- Low back reflection, return loss RL = 45 dB

## ORDERING INFORMATION

### APDTIA4B-2.5G-W-X-X-X-X-X-X

#### Optical matching

**R45:** back reflection -45 dB (SM1 and SM3 fiber)

**R30:** back reflection -30 dB (MM5 fiber)

**RM:** optical matching, +5% larger responsivity

#### Case type

**COAX:** compact coaxial (low duty cycle pulse mode only)

**COAXB:** compact coaxial with a bracket

**COAXH:** compact coaxial with an H-shaped bracket for board mounting

#### Pinout code

**10:** see more details on page 4

#### Fiber type

**SM1:** SM, G.657.A1, [Corning SMF-28 Ultra](#), furcation tubing Ø0.9 mm or **BSM1** Ø0.25mm

**SM3:** SM, G.657.B3, [Corning ClearCurve ZBL](#), furcation tubing Ø0.9 mm or **BSM3** Ø0.25mm

**MM5:** MM, [50/125. OM3](#), furcation tubing Ø0.9 mm

Other type on request

#### Connector type

**FU:** FC/UPC (SM1, SM3, MM5)

**FA:** FC/APC (SM1, SM3, MM5)

**SU:** SC/UPC (SM1)

**SA:** SC/APC (SM1)

**N:** no connector (scissors cut)

Other type: on request

#### Fiber length

**0.5:** 500+/-50 mm

**1.0:** 1000+/-100 mm

Other length on request

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## ABSOLUTE MAXIMUM RATINGS

| Parameter              |                   | Value           | Unit | Conditions     |
|------------------------|-------------------|-----------------|------|----------------|
| TIA supply voltage     | V <sub>CC</sub>   | 4.5             | V    |                |
| APD voltage            | V <sub>APD</sub>  | V <sub>BR</sub> |      |                |
| Reverse current        | I <sub>R</sub>    | 2               | mA   |                |
| Operating temperature* | T <sub>op</sub>   | -40 - +85       | °C   |                |
| Storage temperature    | T <sub>stg</sub>  | -40 - +85       | °C   |                |
| Soldering temperature  | T <sub>sold</sub> | 260             | °C   | Max. 5 seconds |

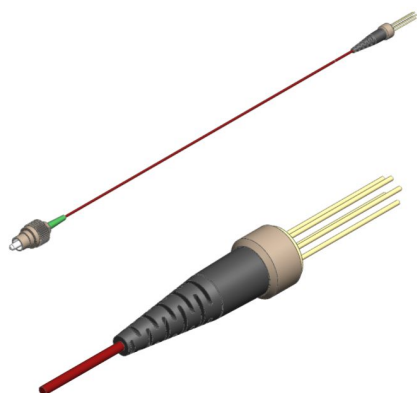
\*Operating temperature is defined by the case temperature. It is necessary to ensure sufficient heat dissipation so that the module's maximum operating temperature is not exceeded. Operation at elevated temperatures reduces the lifetime of the module.

# APDTIA4B-2.5G-W

## ELECTRICAL-OPTICAL CHARACTERISTICS (T = 25 °C)

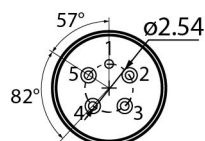
| Parameter                                                          |                  | MIN  | TYP  | MAX   | Unit              | Conditions                                                                         |
|--------------------------------------------------------------------|------------------|------|------|-------|-------------------|------------------------------------------------------------------------------------|
| Operating wavelength                                               | $\lambda$        | 1260 |      | 1640  | nm                |                                                                                    |
| TIA supply voltage                                                 | V <sub>cc</sub>  | 3.0  | 3.3  | 3.6   | V                 |                                                                                    |
| TIA supply current                                                 | I <sub>cc</sub>  | 15   | 20   | 24    | mA                | No load                                                                            |
| Responsivity @ unity gain M = 1                                    | R                | 0.85 | 0.95 |       | A/W               | R45, R30, $\lambda$ = 1550nm, P = 1uW                                              |
| Responsivity @ unity gain M = 1                                    | R                | 0.90 | 1.00 |       | A/W               | RM, $\lambda$ = 1550 nm, P = 1uW                                                   |
| Return loss                                                        | RL               | 40   | 45   |       | dB                | R45, SM1, SM3                                                                      |
| Return loss                                                        | RL               | 25   | 30   |       | dB                | R30, MM5                                                                           |
| Return loss                                                        | RL               | 25   | 30   |       | dB                | RM                                                                                 |
| Breakdown voltage                                                  | V <sub>BR</sub>  | 35   | 46   | 56    | V                 | I <sub>d</sub> = 10 uA                                                             |
| Breakdown voltage temperature coefficient $\Delta V_{BR}/\Delta T$ | $\delta$         | 0.08 | 0.10 | 0.12  | V/C               | T = 25 C                                                                           |
| Dark current                                                       | I <sub>d</sub>   |      | 10   | 50    | nA                | V <sub>R</sub> = 0.9 V <sub>BR</sub>                                               |
| Data rate                                                          |                  |      | 2.5  |       | Gbps              |                                                                                    |
| Low-frequency cut-off                                              | f <sub>c</sub>   |      | 30   |       | kHz               |                                                                                    |
| Optical sensitivity                                                | P <sub>min</sub> |      | -35  | -33.5 | dBm               | BER = 10 <sup>-10</sup> , BR = 2.5 Gbps NRZ, PRBS, ER = 10 dB, $\lambda$ = 1550 nm |
| Output voltage                                                     | V <sub>out</sub> |      | 140  |       | mV <sub>p-p</sub> |                                                                                    |
| Transimpedance                                                     | Z <sub>t</sub>   | 5.7  | 7.7  | 9.0   | kOhm              | Differential (50 $\Omega$ on each output), f = 100 MHz                             |
| Output impedance                                                   | Z <sub>out</sub> |      | 50   |       | Ohm               |                                                                                    |
| Overload                                                           | P <sub>max</sub> | 2    |      |       | dBm               | BER = 10 <sup>-10</sup> , BR = 2.5 Gbps NRZ, PRBS, ER = 10 dB, $\lambda$ = 1550 nm |

# APDTIA4B-2.5G-W



## COAX

### BACK VIEW



### PINOUT

#10

1. Gnd
2. Dout
3. Vcc
4. Vapd
5. Dout

Download more  
information

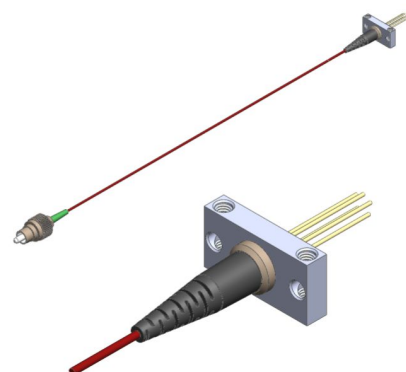
Drawing



3D model

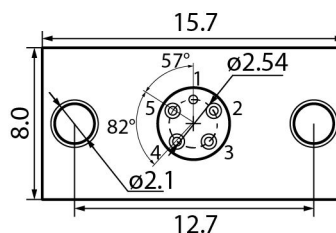


Application Notes



## COAXB

### BACK VIEW



### PINOUT

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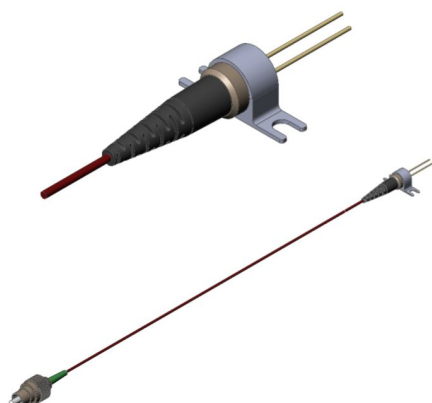
Drawing



3D model



Application Notes



### PINOUT

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Download more  
information

Drawing



3D model



Application Notes

# APDTIA4B-2.5G-W

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Characteristics, data, materials and structures specified in this datasheet are subject to change without notice. Please refer to the latest specification before use of the products.

## Safety and handling cautions

1. Avoid smashing and burning of the module. Avoid storing and using the module in conditions where water, organic solvents or aggressive acids or bases may contact the module or where there is a possibility of exposure to corrosive gases, explosive gases, dust, salinity or other harsh conditions. The module should be disposed as special industrial waste.
2. Exceeding absolute maximum ratings even for a short time can cause permanent damage of the module.
3. The module is sensitive to and can be broken by ESD (static electricity).

## Conflict Minerals Policy Statement

LD4B achieves business objectives and customer needs with social responsibility. We do not support or contribute to the violence and human rights violations associated with the mining of conflict minerals coming from Conflict Regions according to US "Dodd-Frank Act". When possible, our suppliers' conflict mineral statements are reviewed. We do not directly purchase Conflict Minerals from any source and do not knowingly procure any parts and products containing Conflict Minerals from Conflict Regions.

## RoHS Compliance Statement

Restriction of Hazardous Substances (RoHS) directive (Directive 2011/65/EC amended with Directive (EU) 2015/863) is the directive aimed at reducing the harmful environmental impact of waste electrical equipment by restricting the use of known dangerous substances. Based on information received from our supply sources, LD4B hereby states that the banned substances listed in the RoHS directive are not found in the parts and materials used above the threshold level listed other than exceptions approved by the European Commission.

## REACH Compliance Statement

Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH) is a European Union regulation 1907/2006/EC that addresses the production and use of chemical substances, and their potential impacts on human health and the environment. Based on information received from our supply sources, LD4B hereby states compliance of the parts and materials used in manufacturing to REACH regulation. LD4B does not manufacture or import any substances or preparations as defined under REACH.